

Table A2. Critical crop growth periods and depth to which roots of mature crops will extract available water under average unrestricted and restricted conditions.

Crop	Critical Cropping Period	Normal Effective	Min Effective
		Rooting Depth Unrestricted - inches (m)	Rooting Depth Restricted - inches (m)
Agronomic			
Alfalfa	Early Spring and immediately after cuttings	24 (0.6)	18 (0.5)
Corn, grain	15 days prior to and 15 days after silking	36 (0.9)	24 (0.6)
Corn, silage	15 days prior to and 15 days after silking	36 (0.9)	24 (0.6)
Corn, sweet	From silking through ear formation	30 (0.7)	18 (0.5)
Cotton	During and immediately after bloom stage	36 (0.9)	24 (0.6)
Grain, small	Planting and 2 weeks before pollination through head formation	24 (0.6)	18 (0.5)
Pasture grass	At planting and throughout Summer	36 (0.9)	30 (0.7)
Peanuts	Nut enlargement stage	24 (0.6)	18 (0.5)
Sorghum, grain	From boot to flowering stage	36 (0.9)	24 (0.6)
Soybeans	Pod filling stage	30 (0.7)	24 (0.6)
Tobacco	Transplanting, knee high to bloom, during harvest	18 (0.5)	18 (0.5)
Turfgrass	Planting and throughout growing season	6 (0.15)	6 (0.15)
Vegetables			
Asparagus	Crown set and transplanting	24 (0.6)	18 (0.5)
Beans, dry/snap	During and immediately following bloom	24 (0.6)	18 (0.5)
Beans, Lima	During and immediately following bloom	30 (0.7)	24 (0.6)
Beans, soy	During and immediately following bloom	24 (0.6)	18 (0.5)
Beets, sugar	During early growth and root expansion	36 (0.9)	24 (0.6)
Broccoli	Throughout growing season	18 (0.5)	12 (0.3)
Cabbage	Last 3-4 weeks prior to harvest	18 (0.5)	12 (0.3)
Cabbage, Nappa	Throughout growing season	18 (0.5)	12 (0.3)
Carrot	Seed germination, root expansion	18 (0.5)	12 (0.3)
Cauliflower	Throughout growing season	18 (0.5)	12 (0.3)
Celery	Throughout growing season	18 (0.5)	12 (0.3)
Collards	Throughout growing season	18 (0.5)	12 (0.3)
Cucumber	Flowering and fruiting	18 (0.5)	12 (0.3)
Eggplant	Flowering and fruiting	18 (0.5)	12 (0.3)
Greens	Throughout growing season	18(0.5)	12 (0.3)
Leeks	Throughout growing season	18 (0.5)	12 (0.3)
Lettuce	Throughout growing season	24 (0.6)	18 (0.5)
Melons, sweet	At pollination and 2-3 afterwards	36 (0.9)	24 (0.6)
Okra	From bloom through harvest	24 (0.6)	18 (0.5)
Onion	Throughout growing season	18 (0.5)	12 (0.3)
Peas, crowder	From bloom through harvest season	18 (0.5)	12 (0.3)
Peppers	1-2 weeks prior to bloom to 2-3 weeks prior to end of harvest	18 (0.5)	12 (0.3)
Potato, Irish	4 weeks prior to harvest	18 (0.5)	12 (0.3)
Potato, sweet	During rapid root expansion	24 (0.6)	18 (0.5)
Squash, Summer	From bloom through harvest season	24 (0.6)	18 (0.5)
Squash, Winter	From bloom through harvest season	24 (0.6)	18 (0.5)
Tomatoes	1-2 weeks prior to bloom to 2-3 weeks prior to end of harvest	24 (0.6)	18 (0.5)
Fruit and Nut			
Blueberries	Transplanting and from bloom until harvest	24 (0.6)	18 (0.5)
Grapes	Transplanting, and during fruit enlargement	60 (1.5)	36 (0.9)
Peaches	During final swell prior to harvest	60 (1.5)	36 (0.9)
Pecans	During nut set (Apr-May) and nut fill (Aug-Sept)	60 (1.5)	48 (1.2)
Strawberries	Transplanting and from bloom until harvest	12 (0.3)	10 (0.2)

Source: North Carolina Irrigation Guide 2010.